A Gaussian sparse SYK instance for wormhole teleportation on IBM Nighthawk

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Abstract

In response to a question from the authors of [1], we look for an N=8 Gaussian sparse SYK Hamiltonian with about 24 terms that gives a large teleportation signal in both protocols with few two-qubit gates on IBM Nighthawk. We find one K=24 term set that routes to 216/225 two-qubit gates (FIX/SYM), against 309/410 for a typical draw, and passes every chaos, signal and robustness check we fixed in advance. In a FakeNighthawk noise simulation it keeps 25 to 35% more signal than their binary Eq. 11 compiled the same way and 8 and 9 times a typical draw (FIX/SYM); with term order optimized on both sides, FIX keeps +18% and SYM is at best a tie.

Problem. The authors of [1] asked for an N=8 Gaussian sparse SYK Hamiltonian with about 24 terms, a large teleportation signal ΔI in both the symmetric (SYM) and fixed-injection-time (FIX) protocols, and the fewest two-qubit (2q) gates after routing on IBM Nighthawk. Two coupled Sachdev–Ye–Kitaev (SYK) models in a thermofield double (TFD) state model a traversable wormhole: a qubit swapped into the left side at $-t_0$ reappears on the right at t_1 when the coupling $e^{i\mu V}$ between the sides (V a sum of left-right Majorana bilinears) has $\mu < 0$ [2, 3, 4]; small instances have run on hardware [5], with debate on how well they scramble [6]. Sparse SYK keeps K of the $\binom{N}{4}$ four-Majorana terms and stays chaotic above a threshold K [7, 8, 9]. Ref. [1] measures $\Delta I = I_{PT}(\mu{=}-12) - I_{PT}(\mu{=}+12)$, the base-2 mutual information between a reference qubit P and the readout qubit T, at $\beta = 3$ on 10 qubits, for SYM ($t_0 = t_1$) and FIX ($t_0 = 1.8$, t_1 varied), and runs a binary K=10 instance (their Eq. 11) on IBM hardware. The task: Gaussian couplings ($\mathrm{Var}\, J = 1.640625/K$), $K \approx 24$ of 70 terms, large ΔI in both protocols, and few routed 2q gates (our bar: at most 0.70× the typical median).

Result. An *instance* is one term set plus one Gaussian coupling draw. Table 1 lists every instance that passed all gates, plus two controls. The first two share term set d6a76272, the recommended set. The third, 1718_714 on set 2f2d8f4f, passes by the pre-registered rule (list A: FIX +2.8%, SYM +0.8%) but falls below Eq. 11 on held-out lists B and C (FIX −0.05%, −3.6%; SYM −1.4%, −3.1%); its SYM margin is within seed noise, so we call it borderline, not a second full pass. 19277_201 keeps +25–35% / +25–29% more noisy ΔI than Eq. 11 and 8.1 / 9.4 times a typical draw; 155_129 has the smaller margin (FIX +6% on list B). Before the noise test, 23 instances on 14 term sets passed the robustness gates: the cost bar on three seed lists; eight noiseless ΔI peaks (exact and Trotter, FIX and SYM, pair (1,2) and 28-pair mean) all $\geq 1.02\times$ Eq. 11's; at least 45 of 50 draws with 1% coupling noise keep every peak above Eq. 11's; level-spacing ratio $r \geq 0.386$ (Poisson; GOE 0.53; 8 levels) in both parity sectors. One of the 14 sets also beats Eq. 11 under noise on all three lists. Its two instances have $r = 0.44$ / 0.45 (19277_201) and 0.50 / 0.40 (155_129), the borderline instance 0.56 / 0.65, against 0.537 / 0.479 for Eq. 11. The instances cost more 2q gates than Eq. 11; the cheapest certified instance routes to FIX 197 on another set (K=25, which fails the noise test). New beyond [1]: they locate the Gaussian threshold ($K \approx 23$) and run binary Eq. 11; here a specific Gaussian K=24 instance with FakeNighthawk-routed circuits keeps more simulated noisy signal than Eq. 11 in canonical term order (FIX also with optimized order).

Instance	K	2q FIX / SYM	noiseless ΔI	noisy ΔI FIX	noisy ΔI SYM	vs Eq. 11 (A–C)	× typical
19277_201 (set d6a76272)	24	216 / 225	0.33 / 0.28	0.12 ± 0.005	0.096 ± 0.0006	+25–35% / +25–29%	8.1 / 9.4
155_129 (set d6a76272)	24	216 / 225	0.27 / 0.25	0.097 ± 0.004	0.087 ± 0.0007	+6–14% / +14–18%	6.8 / 8.5
1718_714 (set 2f2d8f4f), borderline	24	208 / 250	0.25 / 0.26	0.092 ± 0.002	0.075 ± 0.0003	−4 to +3% / −3 to +1%	6.4 / 7.3
Eq. 11 (binary, control)	10	175 / 215	0.21 / 0.20	0.089 ± 0.0009	0.074 ± 0.001	ref.	6.3 / 7.2
Typical K=24 (median, n=20)	24	309 / 410	0.060 / 0.079[a]	0.014	0.010	−84% / −86%	1 / 1

Table 1: Instances that pass every gate, and controls. 2q: routed cz count, FIX / SYM; the lower median over 20 routing seeds, which is also the shipped circuit's count. Noiseless ΔI: routed circuit with the fitted TFD at t_1, bits. Noisy ΔI: FakeNighthawk gate and readout noise, median over the central 5 of 20 routing seeds (sorted by 2q, depth, seed) of seed list A; ± is half the range of the medians over seed lists A, B, C. "vs Eq. 11": range over lists A, B, C, each against Eq. 11 on the same list; ratios use unrounded medians. "× typical": list A. Typical: 20 uniform K=24 draws, list A; [a]over the 17 draws with all 5 seeds; 2q is the 200-draw median.

Method. Term sets: 5×10^8 uniform K=24 term-set draws ranked by a cheap predicted cost, 30,842 kept, the top 200 routed on FakeNighthawk. Couplings: up to 2×10^7 i.i.d. Gaussian coupling draws per set, screened on chaos (r, spectral form factor, OTOC), the eight ΔI peaks and Trotter error, then a 720-start variational TFD fit (fidelity $F \geq 0.96$); the couplings are selected draws, never continuously optimized. Circuits: one Lie–Trotter step (FIX) or

Suzuki-2 step (SYM), the Jordan–Wigner qubit map S2 of [1] for both protocols, canonical greedy term order, per-block pytket 2.18.5 GreedyPauliSimp and FullPeepholeOptimise, then Qiskit 2.5.2 at optimization level 3 on FakeNighthawk. Noise: Aer density-matrix simulation from FakeNighthawk (qiskit-ibm-runtime 0.49.0) on the shipped circuits, two-qubit tomography of P and T.

Verification. Known answers first: Eq. 11 gives $r = 0.537163$ / 0.478812 against the printed 0.537 / 0.479; Eq. 11's exact pair-(1,2) ΔI peaks, 0.2127 (SYM) and 0.2762 (FIX), match Fig. S1 of [1] (0.212 and 0.276, read to ± 0.003); the Gaussian transition points of their Table S1 come out at $K_1 = 22$ [21, 24] against 23 and $K_2 = 20$ [18, 69] against 23 (consistent, weakly constrained). Only verdicts of hash-frozen verifier code count; seed list A searched, B certified, C held out from the search; C also gates the cost check, and the robust set's noisy ordering holds on all three. A blind re-derivation from the shipped files alone reproduced every number needed to run the shipped circuits on hardware to 2.8×10^{-7} (two text errors found and fixed, no number errors). Two independent adversarial reviews, kept with the code, found no blocking issue.

Limits. (i) FakeNighthawk is a simulated proxy for the device. With idle noise added, 19277_201 keeps +21% / +22% (0.078 / 0.066 vs 0.064 / 0.054) and 155_129 ties Eq. 11 on FIX (0.0642 vs 0.0644) and keeps +10% on SYM (0.059). With every FakeNighthawk error rate scaled up, 155_129 FIX falls to Eq. 11 at about 1.5 times the rates, while 19277_201 stays 16–25% above it up to 1.75 times. (ii) With term order optimized on both sides, Eq. 11 SYM routes to 134–135 2q; 19277_201 FIX keeps +18%, but SYM loses in two of three order searches (−10%; −10% / −11% on lists A / B) and ties in the third (+1% / −7%). (iii) Selection and multiplicity: the instances are about 1 in 1.2–1.8×10^{13} screened coupling draws; 19277_201 clears some gates narrowly (OTOC decay time 2.94 against the bar 2.95, F 0.965 vs 0.96); the second robust instance came from a search on the set that had just passed; the final search sent 6 instances on 5 sets to the verifier and 1 passed, the borderline one, by +0.8% on SYM, a margin within the 8–14% per-seed spread. (iv) Eq. 11's noiseless Trotter peaks (FIX / SYM) are 0.2301 / 0.2147 under a term order and an H_L-only first stage that we reconstructed; the literal reading of [1] gives 0.2246 / 0.2130. Both sides use the reconstruction. (v) Ref. [1] ran SYM in its S3 map and printed 150 / 282 2q for Eq. 11; our pipeline in S3 gives 222 / 274. With S3 on both sides (report only) both robust instances still pass (19277_201 +30% / +28%, 155_129 +19% / +15%).

Code. `https://github.com/oharalabsai/syk-gaussian-n8`. `uv sync && uv run verify` re-derives the 2q and noiseless ΔI columns of Table 1 for all three instances and Eq. 11, with Eq. 11's known answers and every instance's noiseless numbers, in about a minute on a laptop; `uv run noise <instance> --protocol FIX` gives the FakeNighthawk noisy ΔI of one shipped circuit pair (the table's noisy columns are medians over five routing seeds).

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AI contribution. The authors are responsible for all content. An AI agent harness ran the search, simulations and first drafts under human direction; every number comes from a frozen verifier or a logged job. No AI is an author.

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